Reducing permutation degree through a commuting action

Human Compression project: working common argument

A free transitive group action supplies coordinates without selecting an identity point intrinsically: it is a torsor. Choosing one point in each orbit of a semiregular permutation group makes the action of its centralizer particularly explicit. A quotient of these coordinates can reduce the permutation degree while preserving the entire commuting group, including its marked subgroup intersections. The quotient need not preserve the original geometry; the assertion concerns a faithful representation of the same abstract group.

Lemma 1. *Let $H \leq \mathrm{Sym}(\Omega)$ be transitive and let $D \leq C_{\mathrm{Sym}(\Omega)}(H)$. Then D is semiregular.*

Proof. If $d \in D$ fixes x, then it fixes every $h(x)$, because $dh = hd$. Transitivity implies that d fixes Ω pointwise. Since D is a permutation group, $d = 1$. □

Theorem 2. *Let a finite group D act semiregularly on a nonempty finite set Ω, and let b be its number of orbits. Suppose $K \leq D$ has trivial core,*

$$\mathrm{Core}_D(K) = \bigcap_{d \in D} d^{-1}Kd = 1.$$

There is a faithful permutation representation of $C_{\mathrm{Sym}(\Omega)}(D)$ on $b[D : K]$ points. Every transitive subgroup H of this centralizer remains transitive in the new representation. If $b > 1$ and $[D : K] > 1$, the new action preserves a nontrivial system of b blocks of size $[D : K]$.

Proof. Choose one point x_j in each D-orbit, indexed by a set B of size b. Semiregularity identifies Ω with $D \times B$ by $(d, j) \mapsto d(x_j)$. The action of D in these coordinates is left multiplication in the first coordinate.

A permutation h commutes with D if and only if there are $\sigma \in \mathrm{Sym}(B)$ and elements $a_j \in D$ such that

$$h(d, j) = (da_j, \sigma(j)).$$

Indeed, its values at the x_j determine all other values by commutation. Conversely every displayed map is a bijection and commutes with left multiplication. Composing two such maps again gives a map of the same form. This is the usual wreath description of the centralizer, with the right regular action in each coordinate.

Now let $\Delta = K\backslash D$ be the right D-set of right cosets, and define the action on $\Delta \times B$ by

$$\widehat{h}(Kd, j) = (Kda_j, \sigma(j)).$$

It is well defined, and the same composition formula shows that $h \mapsto \widehat{h}$ is a homomorphism. If $\widehat{h}$ fixes every point, then $\sigma = 1$, and every a_j acts trivially on $K\backslash D$. For any $a \in D$, this last condition is equivalent to $dad^{-1} \in K$ for every $d \in D$, hence to $a \in \mathrm{Core}_D(K)$. Thus all $a_j = 1$, proving faithfulness.

The surjection $(d, j) \mapsto (Kd, j)$ is equivariant for every commuting h. Therefore a transitive H remains transitive on its image. Each $\Delta \times \{j\}$ maps onto $\Delta \times \{\sigma(j)\}$, proving the block assertion. All changes of the chosen orbit representatives give conjugate descriptions of this construction. □

Corollary 3. *In Theorem 2, replace a marked subgroup $(H;(s_i)_{i\in I})$ by $(\widehat{H};(\widehat{s}_i)_{i\in I})$. Generator orders, product orders, irredundance and every marked subgroup intersection identity are unchanged. In particular string C-group representations retain their rank and intersection property.*

Proof. The representation restricts to an isomorphism from H to its image. It sends $\langle s_i : i \in J\rangle$ onto $\langle \widehat{s}_i : i \in J\rangle$ for every $J \subseteq I$. Injectivity implies that images commute with intersections of subgroups. These facts give each assertion, including all pairs of subsets in the C-group intersection property. □

Corollary 4. *Suppose a transitive group $H \leq \mathrm{Sym}(\Omega)$ centralizes a dihedral group D of order $2d$, where $d \geq 3$. Then H has a faithful transitive permutation representation of degree $|\Omega|/2$. If H is a string C-group of rank s and $|\Omega|/2 = m \geq 7$, the transitive rank bound for proper subgroups of symmetric and alternating groups gives*

$$s \leq m/2 + 1.$$

Proof. Lemma 1 makes D semiregular. Take K to be the stabilizer of a vertex in the usual action of D on a regular d-gon. This action is faithful for $d \geq 3$, so K is core-free and $[D : K] = d$. Theorem 2 gives degree $bd = |\Omega|/2$; Corollary 3 preserves the C-group representation.

If $b > 1$, the new action is imprimitive, so its image is neither S_m nor A_m. If $b = 1$, the centralizer of the regular D-action has order $|D|$. A subgroup of it transitive on $|D|$ points must be the entire centralizer. Thus $H \cong D$ has order $2m$, which for $m \geq 7$ is smaller than $|A_m|$. In both cases the stated transitive rank theorem applies. That theorem is a retained premise here, with its complete proof in the accepted common theory for Topic B; it is not reproved by this note. □

For example, let Γ be a string C-group of rank r on $2m$ points, and let a subgroup obtained by deleting two marked generators be transitive and commute with two noncommuting involutions. These involutions generate a dihedral group of order at least six. If $m \geq 7$, the subgroup has rank $r - 2$, so the corollary gives

$$r \leq m/2 + 3 < m.$$

Consequently this configuration cannot occur when $r \geq m$. The conclusion requires actual transitivity of the subgroup; an intransitive action does not satisfy the hypothesis.

Attribution and scope

The centralizer of a regular action and its wreath description are standard permutation group arguments. Cameron, Fernandes, Leemans and Mixer use the dihedral reduction twice in Section 3.2 of *Highest rank of a polytope for A_n*, arXiv:1605.09173v1 (2016), pages 6–7. The general core-free formulation above makes the reusable hypothesis and faithful quotient map explicit. No historical priority is claimed for it.

This working note does not recover that paper's main alternating-group rank theorem. In particular its intransitive cases, two-fracture arguments, finite exceptions and lower bounds remain separate obligations. No whole original, compression factor, conjecture answer or Lean formalization is claimed. A later application must acquire every common theorem it uses; the smaller permutation action is not a discount in manuscript length.